

Video Transcript

Video 1: Introduction (04:34)

As it turns out, the days of one type of database are long gone. Now, there's a multitude of them, and in this section, we're going to be looking at a handful. We'll start off with the traditional. The relational database management system we will then look at a document, a great and flexible model to use and expand and stretch on. Then we'll look at key value pair stores, really even easier to use. A great, a great design change when it comes to the type of databases that you can have. Then we'll look at some of these scalable models, the ones that are distributed and can grow a tremendous volume. The ones that learn from a lot of the projects that were done 20 years ago when it comes to scale and being able to truly in a decentralized way, in a horizontal way, meaning by simply throwing hardware at them, can address this tremendous growth.

We'll end up by looking as well at some of the new online offerings where you're really not concerned about how that implementation is done. You're transferring that responsibility to the cloud provider and you're simply using that resource, a very fast and innovative way to do things. Now, because all of this will be done in a multitude of platforms. Some of them are written in C, some of them are written in Java, some of them are written in other languages. It would be tremendously difficult to do this all by hand.

So, because of it, we will be utilizing containers quite a bit and this will give you some great knowledge to walk away with because this is exactly the same way in which this is done when it comes to industry. So, now let's go ahead and jump into databases. As we are well aware, the world is generating a tremendous amount of data. In fact, you see slides such as the following one, emphasizing the volume of data that's going to be produced every day by autonomous vehicles. And interestingly, this projection falls short. I've seen results of a single engine that can generate gigabytes on its own per minute. So, you can see there that there's a tremendous amount that is going to be generated that is being generated at the moment and that we're really still instrumenting. If you take a look at this slide here at this picture of a city, you can see four major sources of data. You can think of the transit systems, the trains, the buses and many other types of transportation. Same thing could be said of the police. Energy as a very

sophisticated set of apparatus and instrumentation that power cities. You can imagine the amount of data that we might get out of that. And we could say the same thing about health.

And really this is taking place everywhere. In fact, farming is one of the most rapidly transforming industries there is. And when we think about our systems, the systems that power the applications we use today, they look very much like this and they themselves are generating tremendous amounts of data. You can think about the data that's simply the operation of these devices of these components generates in addition to all of the application logic and business applications we put on top of it. Now, we tend to think at times, and you will hear us speak of three tiers, and this is really a practical handle where you have a client, you have a server or something that has some logic, you have some components, and then you have a data store.

Now, you might also hear of the full stack. This is another convenient way to think about applications. You have some front end, you have some back-end logic, something that computes in the background, you have some components, and you have a database, and we call all of this the full stack. However, they still look like the former, much richer, much more diverse, but this is a convenient way to speak about them. Now all of this, both the natural world, both the systems that we built are all generating data and going into data stores. And so, we will be looking at the major building blocks that exist when it comes to data stores.

And you can see here the ones that we will be covering, we have MySQL, Mongo, Redis and Cassandra. Each of these corresponds to a major functional grouping or vertical of types of functionality that you will see in databases. And so, we will be looking at each one of these in much more detail.